Shivansh Kumar P Full Stack AI Engineer

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SUMMARY

Full Stack AI Engineer with 4+ years of experience building production-grade AI applications, scalable backend services, and retrieval-augmented generation (RAG) systems. Experienced in distributed systems, vector search optimization, and ML model deployment using XGBoost and LightGBM. Delivered measurable improvements in API scalability, retrieval performance, and model accuracy across production applications.

WORK HISTORY

Full Stack AI Engineer, *Unibloom* 06/2025 – Present | Sweden/Remote

- Architected and deployed a production-grade web application and AI services using **Next.js and Bun**, applying Domain-Driven Design **(DDD)** principles; reduced AI service cold-start latency by 30% through runtime migration.
- Developed and deployed **XGBoost and LightGBM** models for carbon emissions forecasting, reducing **RMSE by 15%**; served inference through **FastAPI** on **Google Cloud Run** with automated **GitHub Actions CI/CD pipelines**.
- Validated carbon emissions forecasting models against historical baselines using out-of-sample holdout datasets to assess **generalization and quantify prediction error**.
- Optimized ML inference infrastructure on **Google Cloud Run** by tuning **CPU and memory** allocations and configuring autoscaling to balance compute costs with **peak-load performance**.
- Designed and optimized a hybrid retrieval pipeline using PostgreSQL with pgvector and **AI tool integrations**, reducing retrieval **latency by 50%** and improving answer verification.
- Built an automated **Python-based web scraping** and **data extraction pipeline** to process **JavaScript-rendered** web pages into structured Markdown for **LLM ingestion**.
- Integrated **Stripe payment** infrastructure for one-time purchases and **recurring subscriptions**, implementing **error handling** for **reliable payment processing**.

Software Engineer, *Argenbright Innovation Lab India Private Limited* 09/2022 – 03/2025 | Bengaluru

- **Improved the maintainability** of a distributed ecosystem of 40+ microservices built with Node.js and NestJS by applying Domain-Driven Design (DDD), reducing feature delivery cycles by 30%.
- Identified database bottlenecks through **k6 load testing** and optimized **MySQL and PostgreSQL** queries, reducing query latency by **40% across high-traffic endpoints.**
- Designed and optimized **REST and GraphQL APIs** using **Node.js** to **support 12,000+ concurrent users** while maintaining **99.9%** uptime for core business services.

PROJECTS

Unibloom LCA, *AI-powered Life Cycle Assessment (LCA) tool for carbon emission forecasting*

- Built a carbon emissions forecasting service using XGBoost and FastAPI, deployed on **Google Cloud Run with automated GitHub Actions CI/CD workflows**.
- Developed an automated data ingestion pipeline using **Crawl4AI and Tinyfish Search** to **extract web content and index relevant data for model training** and inference.

Unibloom Switch, *AI powered Data accumulator on sustainability goals.*

- Designed a **retrieval-augmented generation** (RAG) architecture with tool-calling capabilities to power **AI agents** that provide sustainability goal recommendations.
- Performance: Developed **semantic search** on massive databases, delivering **70% faster answers** for internal AI agents.
- Optimization: Decreased backend latency and removed context window bottlenecks for **AI agent workflows**.

SKILLS

Language & Architecture
TypeScript, SQL, GO, Python, Node.js, NestJS, NextJs, FastAPI, Microservices,

AI & Data Engineering
RAG, AI Agents, Tool Calling, Embeddings, Semantic Search, XGBoost, LightGBM, PyTorch, Scikit-learn, TabPFN

Databases & ORM
MySQL, Embeddings, PostgreSQL, PGvector, Prisma, Data Ingestion Pipelines

Cloud & MLOps
Google Cloud Platform (GCP), Cloud Run, AWS, CloudFront, Appsync, Lambda Functions, SQS, SNS, Vercel, Supabase, GitHub Actions, CI/CD

EDUCATION

Bachelor of Engineering - BE, *Sri Sairam College of Engineering* 06/2022 | Bangalore

Courses

- Computer Science & Engineering